



ENGLISH

Datasheet

RS PRO Epoxy Glass FR-4

Stock No: 2035179



Technical Specifications:

Colour	Green Black (carbon free)
Sheet Size	420 x 297mm
Maximum Temperature	130°C
Flammability Rating	UL94 V-0
Electrical Strength Condition A	800 V/mil

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Features & Benefits:

- Engineered to provide NEMA grade FR-4 properties
- Woven glass fabric combined with epoxy resin system
- Certifiable to MIL-I-24768/27, Type GEE-F

General Physical Properties	Units	Value
Specific Gravity	-	1.85
Rockwell Hardness	M Scale	98
Moisture Absorption	%	0.1
Flexural Strength LW CW	Psi	65,000 52,000
Tensile Strength LW CW	Psi	40,000 32,000
Izod Impact Strength LW CW	Ft-lb/in Notched	7.9 7.3
Compressive Strength Flatwise	Psi	66,000
Bonding Strength	Lb	2,300
Shear Strength (Perpendicular)	Psi	21,500



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ULTRATRAC® H950 Rigid Laminate

HAYSITE

Outstanding Characteristics & Typical Applications

General	Test Method	Units	Value
NEMA Standards	-	-	GPO-3
ASTM Standards	-	-	D-1532-67T
Government Specifications	-	-	MIL-I-24768/6
Color, Standard	-	-	Red or White
Thickness, Min. - Max.	-	mm	2.4 to 50.8
PHYSICAL			
Barcol Hardness	Barcol	Barcol Scale	62
Specific Gravity	D-792	-	1.80
Density	-	Lbs / in ³	0.065
Water Absorption	D-229	%	0.20
UL Flammability, File No. E81893	UL94	Class	94V-0
Flame Resistance	Ignition Time D-229-11	Burning Time Seconds	130 33
Radiant Panel	E-162	Flame Spread	5.0
Smoke Density @ 4.0 minutes, Flaming	E-662	Optical Density	0
Tunnel Test 6.4mm thick	E-84	Flame Spread	<25
Temperature Class*	-	°C	160
MECHANICAL			
Tensile Strength	D-638	PSI	11,000
Flexural Strength	D-790	PSI	25,000
Flexural Modulus of Elasticity	D-790	PSI	1.50 x 10 ⁶
Compressive Strength	D-695	PSI	30,000
Bond Strength, 1/2" Thick	D-229	PSI	1,400
Shear Strength	D-732	PSI	14,000
Impact Strength, Izod, Edgewise	D-256	Ft.lbs./In.notch	9.50
ELECTRICAL			
Dielectric Strength, 1. Short Time in oil 1.6mm thick	D-149	VPM kV/mm	525 20.67
Dielectric Strength, = Step-by-Step in Oil	D-149	KV	55.0
Arc Resistance	D-495	Seconds	194
Comparative Track Index	CTI	Seconds	600+
Incline Plane Track Resistance 6.4mm thick	D-2303	Minutes	>1000
Dielectric Constant @ 60 Hz	D-150	-	5.20
Dissipation Factor @ 60 Hz	D-150	-	0.06

ULTRATRAC® H950 is a high performance Polyester Glass Rigid laminate which meets or exceeds all specifications for NEMA GPO-3.

ULTRATRAC® H950 is recognized by Underwriters Laboratories as a UL 94V-0 flame resistant material that also offers exceptional arc & track protection.

Testing at an independent laboratory shows H950 to be non toxic as defined by Military Specification MIL-M-14H.

Excellent flame resistance & low smoke generation are also valuable characteristics of this material.

ULTRATRAC® H950 carries a UL thermal recognition of 160°C electrical & 150°C mechanical.

Applications are typical for GPO-3 materials, but also include components for transportation where flame, smoke, & toxicity are of great concern.

*Approximate Temperature Class is listed. Actual Classification depends on end use & location of the material in a complete system.

Unless otherwise noted, all properties are based on 1/8" thick test samples.

Manufacturers Statement

Unless otherwise indicated, all properties published are based on tests performed on standard ASTM test samples and according to ASTM test methods. Values shown are for test samples made from production materials and they are believed to be conservative. No warranty is to be construed, however. In fabricated or molded form, parts may vary considerably from this standard test data. Where specific or unusual applications arise, tests should be made on actual parts, and test procedures agreed upon between Haysite Reinforced Plastics and the customer.



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DEC. 2009

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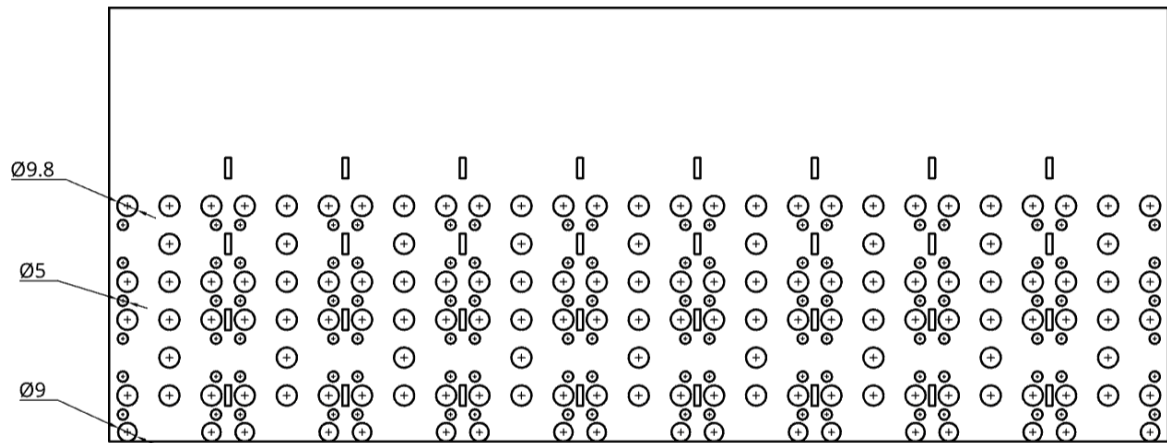
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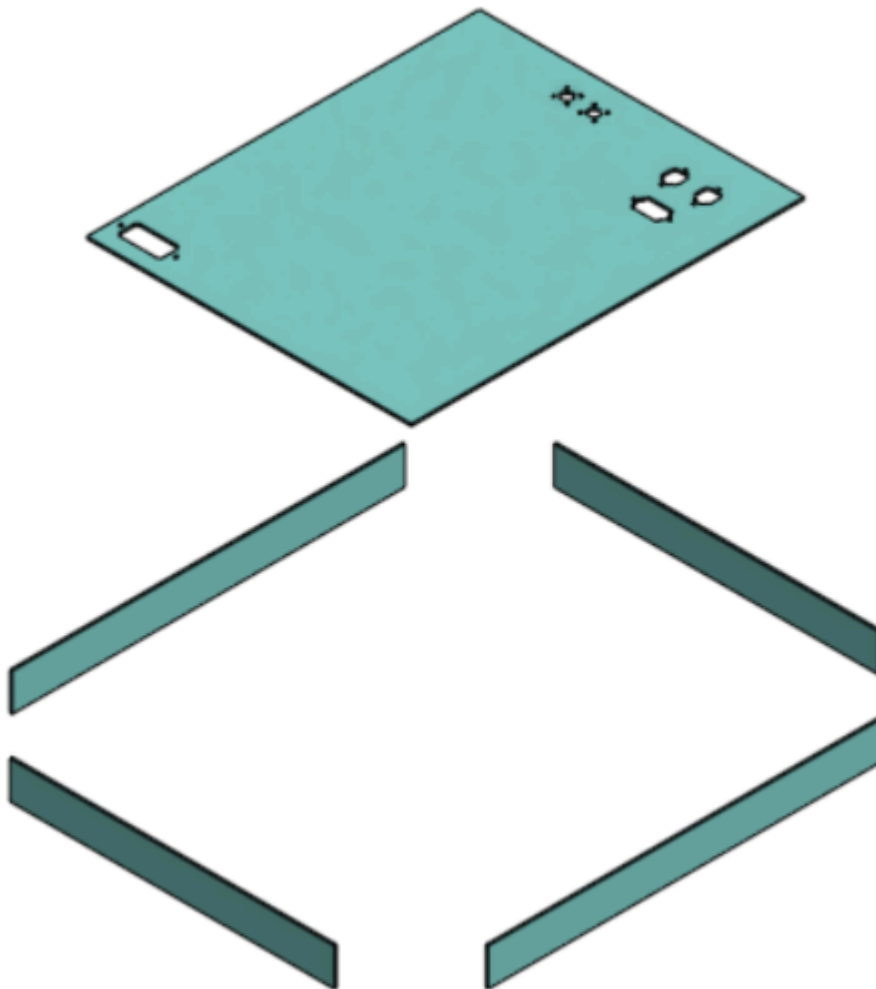
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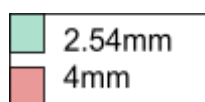
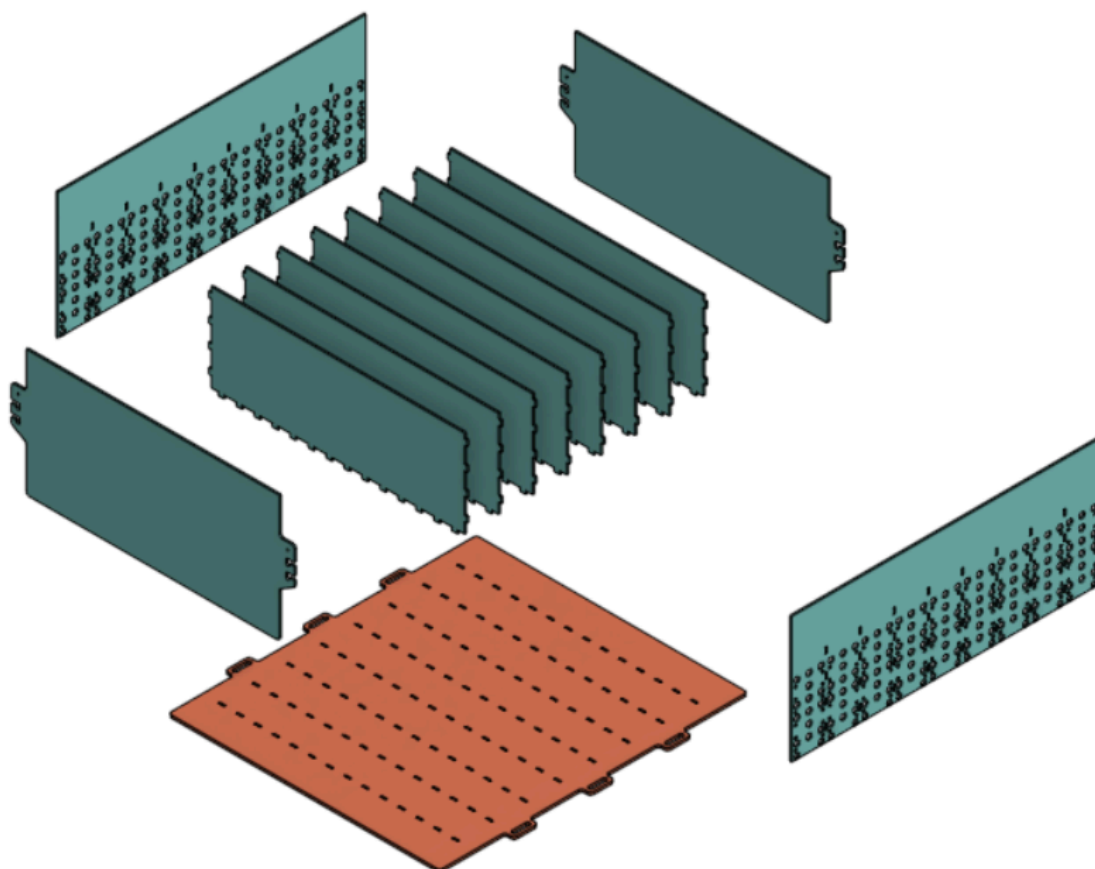
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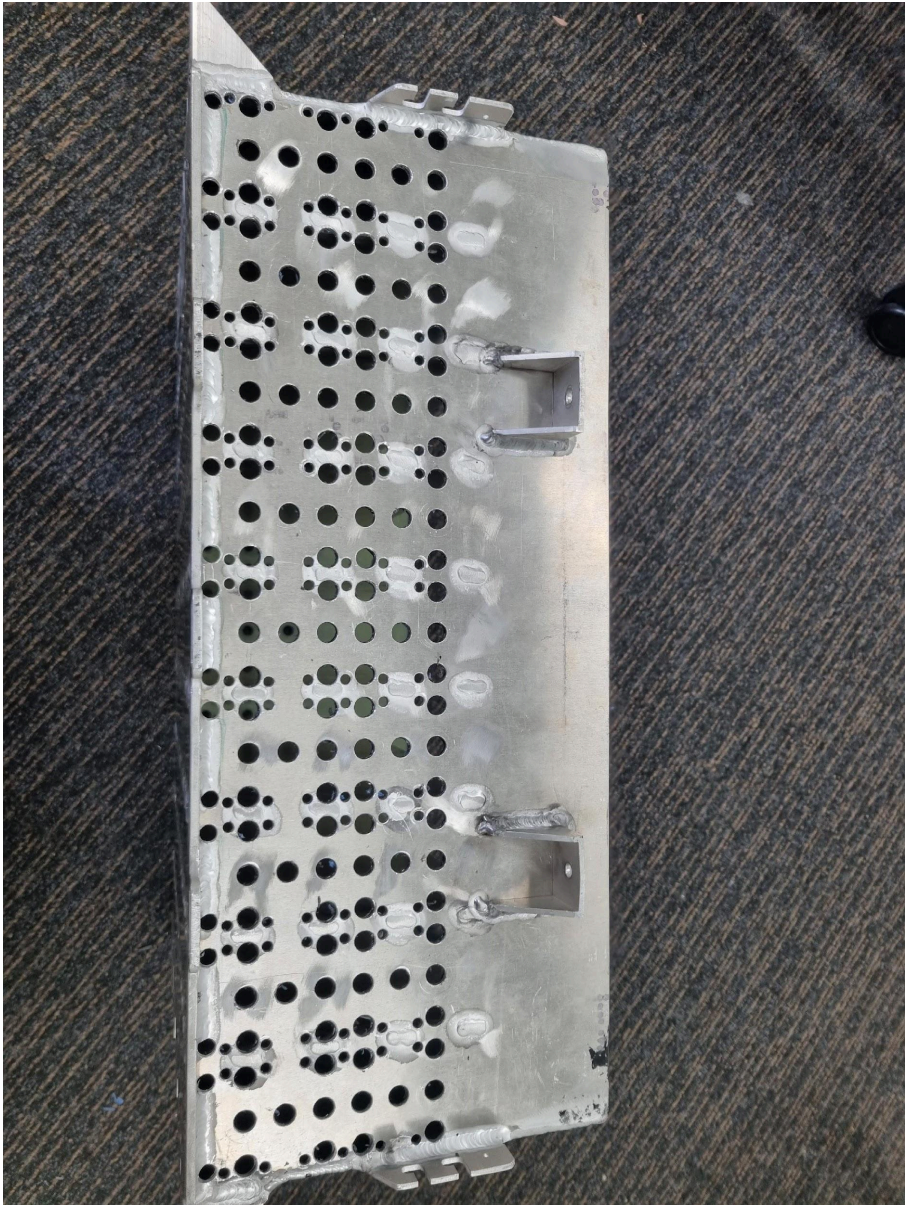
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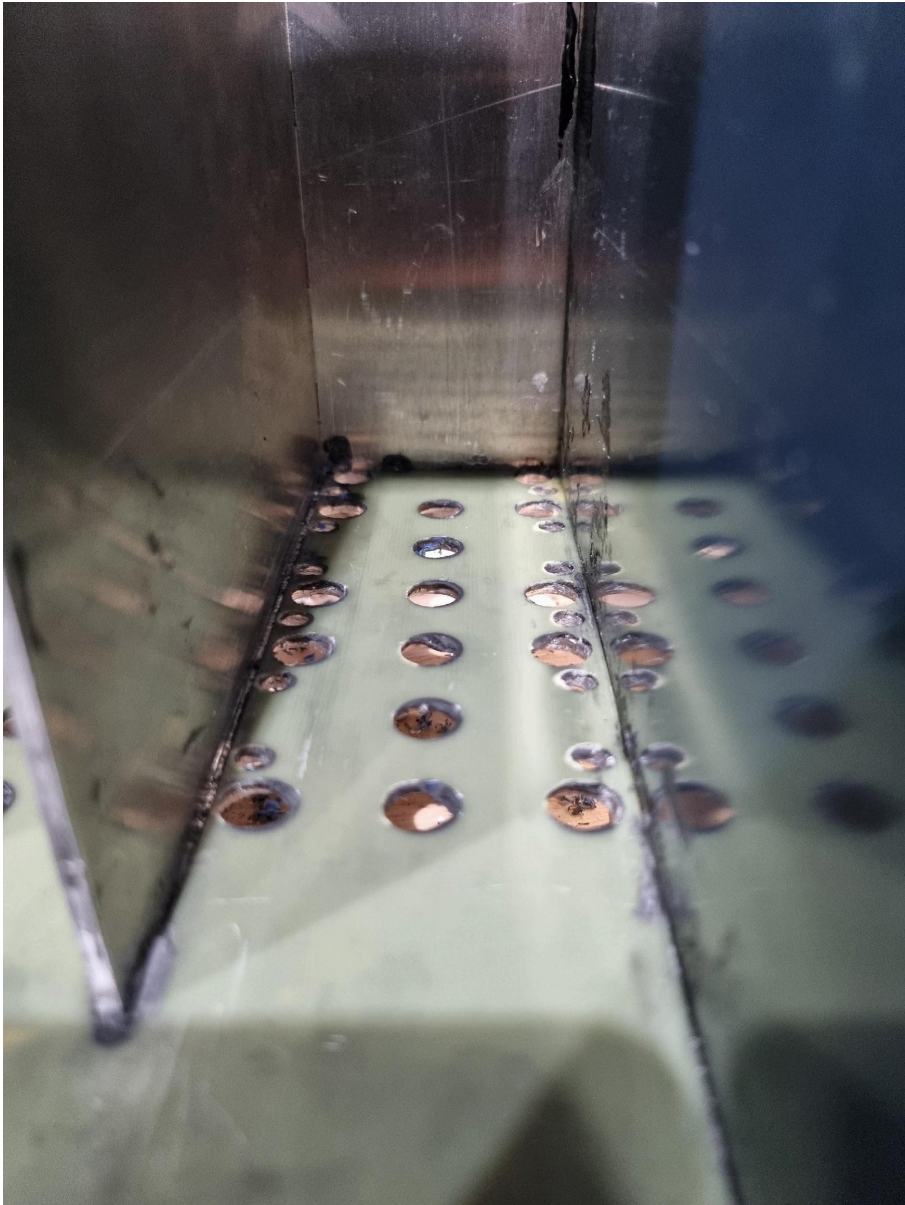


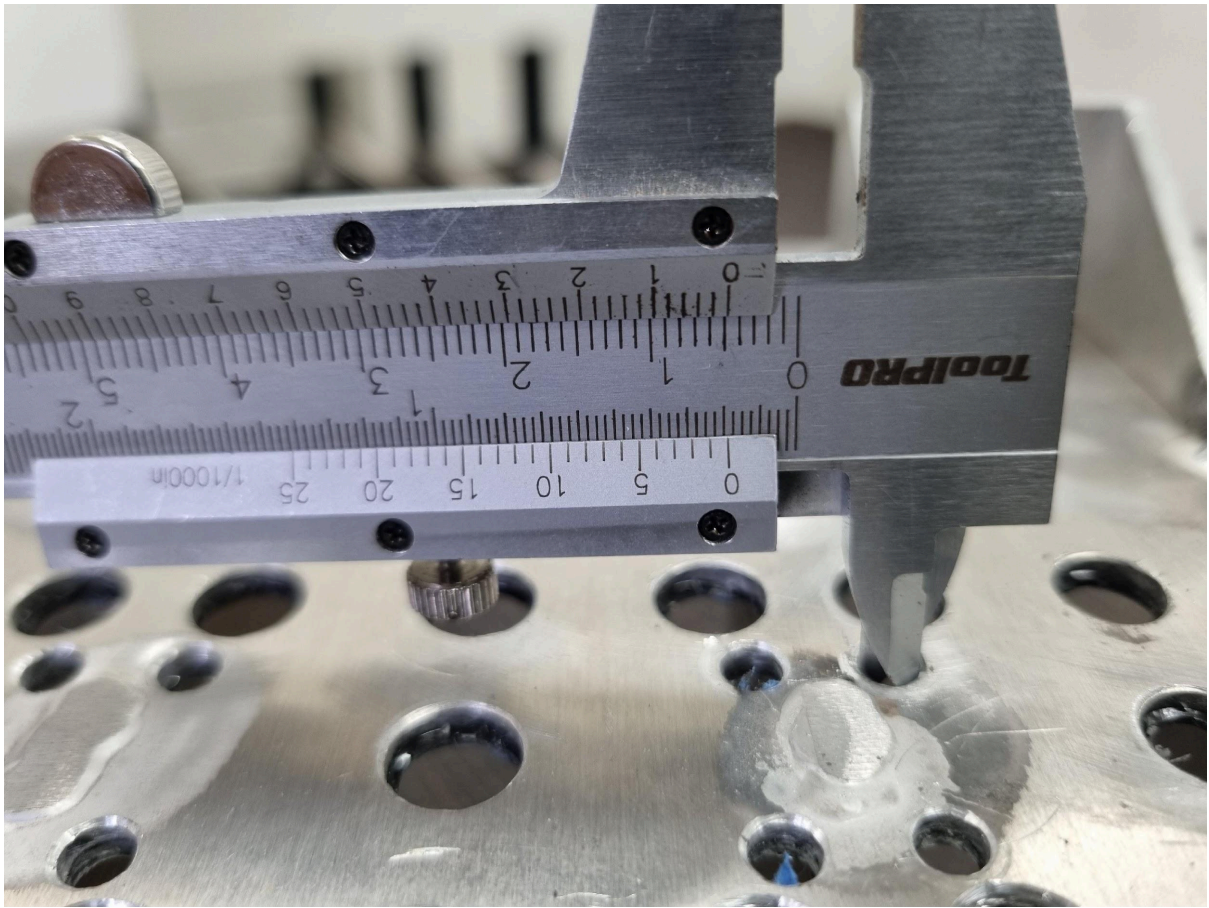
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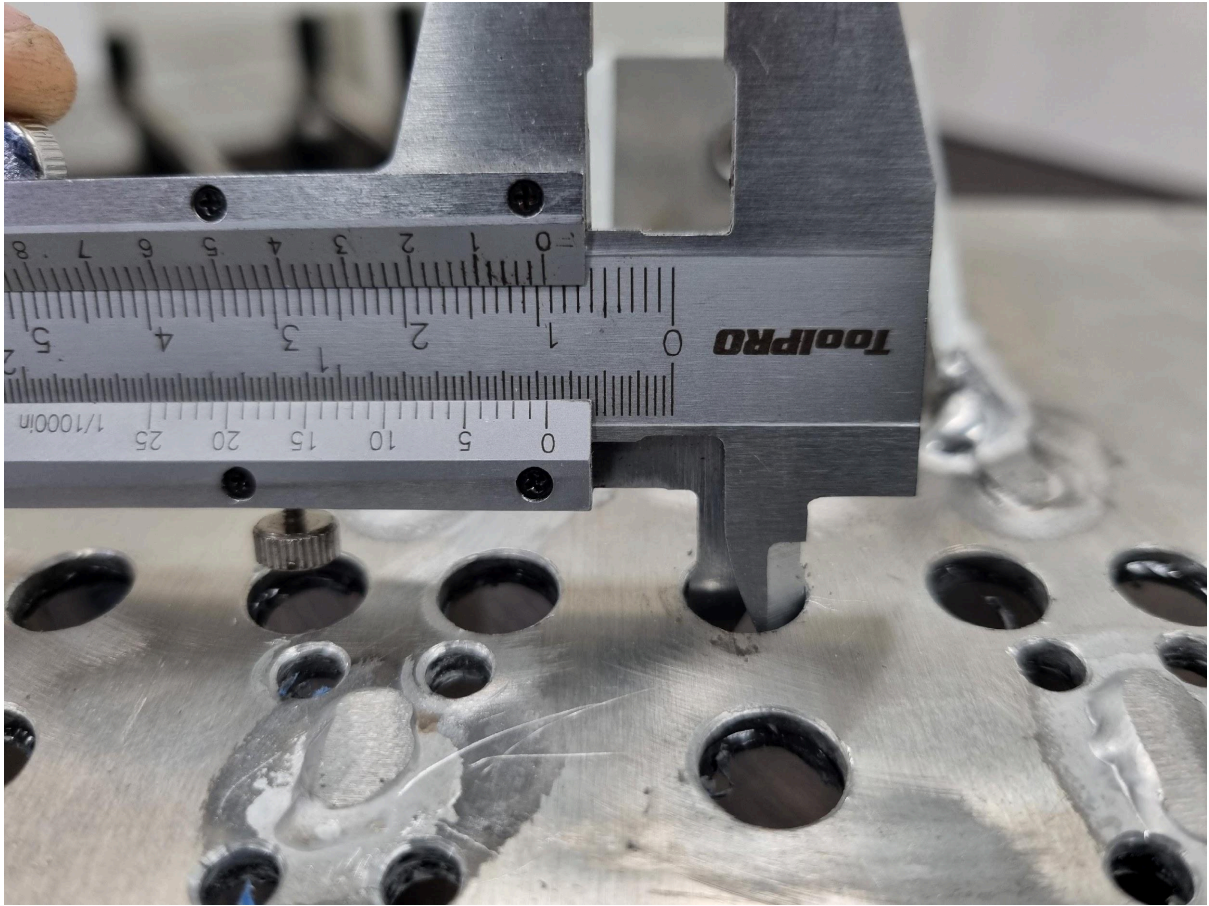


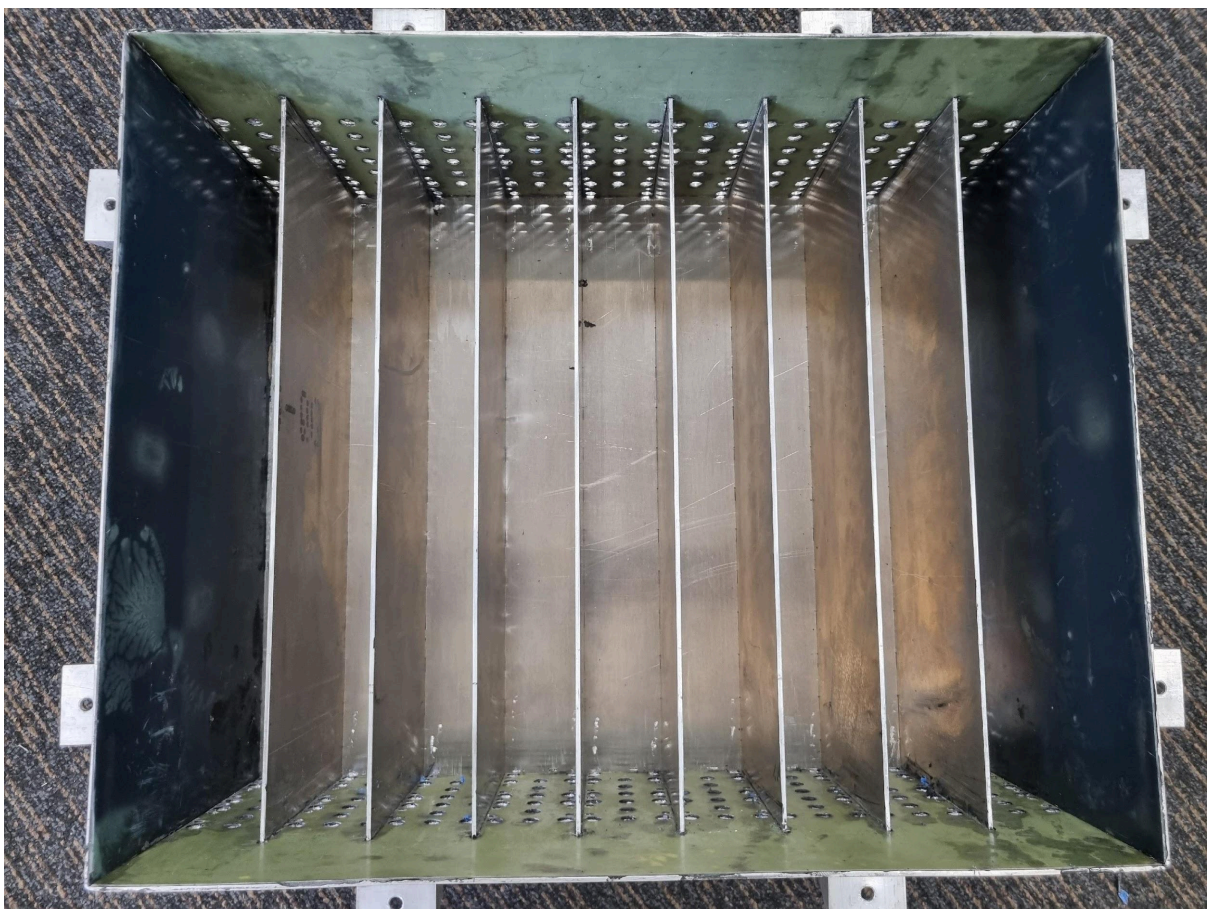
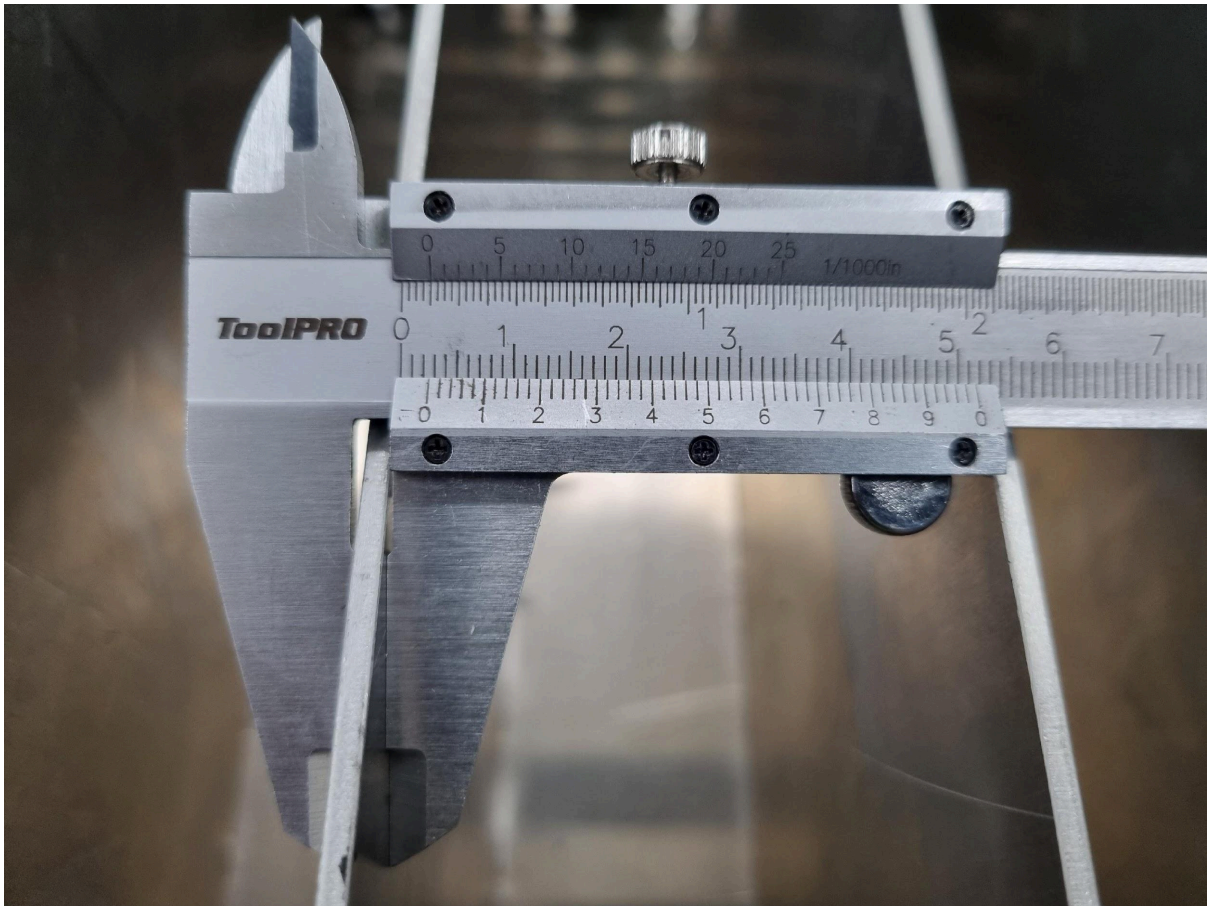












Bolt Details:

Size; M6

Grade; 12.9

Type; Socket Head Cap Screw

No. of Bolts; $N_b = 8$

Tensile Strength; $S_t = 1220 \text{ MPa}$

Tensile Stress Area; $A_t = 20.1 \text{ mm}^2$

Load Details:

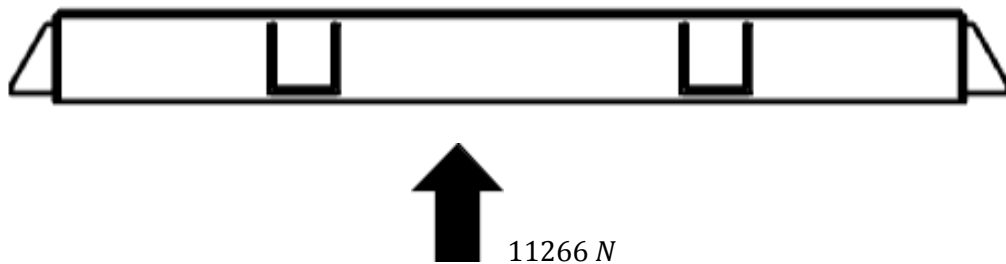
The Max Load Experienced is 20 G in the vertical direction.

Segment Mass; $m_s = 6.38 \text{ kg}$

No. of Segments; $N_s = 9$

Total Segment Mass; $m_t = 57.42 \text{ kg}$

Load; $F = 20 * 9.81 * m_t = 20 * 9.81 * 57.42 = 11266 \text{ N}$



$$\sigma = \frac{F}{A_t * N_b}$$

$$\sigma = \frac{11266}{0.000201 * 8}$$

$$\sigma = 7006219 \text{ Pa}$$

$$\sigma = 7.01 \text{ MPa}$$

$$F_s = \frac{S_t}{\sigma}$$

$$F_s = 174.04$$